MATH 101 Final Exam - Practice 2

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Complete and show all work on a separate sheet of paper. Record final answers on the answer sheet. Give exact answers and simplify completely. Only a basic four-function calculator may be used.

Answer key follows the answer sheet. Attempt the test before checking it.

1. Find (a) domain; (b) hole, vertical asymptote, and horizontal asymptote; (c) both intercepts.

$f(x) = \frac{(x+3)(x+4)}{(x-6)(x+3)}$

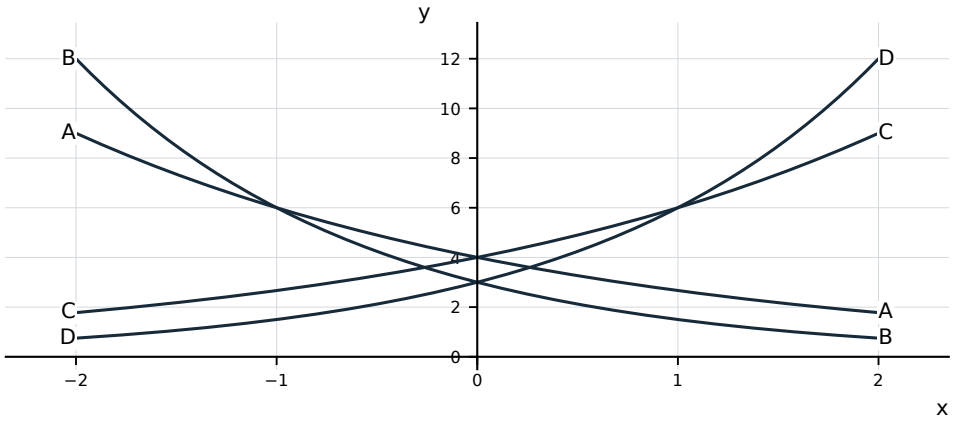
2. Match each function to graph A, B, C, or D. Each letter is used once.

$(a)\quad 4(\frac{3}{2})^x$

$(b)\quad 4(\frac{2}{3})^x$

$(c)\quad 3(2)^x$

$(d)\quad 3(\frac{1}{2})^x$



3. Which pair satisfies h(x) = f(g(x))?

$h(x) = (4x+4)^2 + 5$

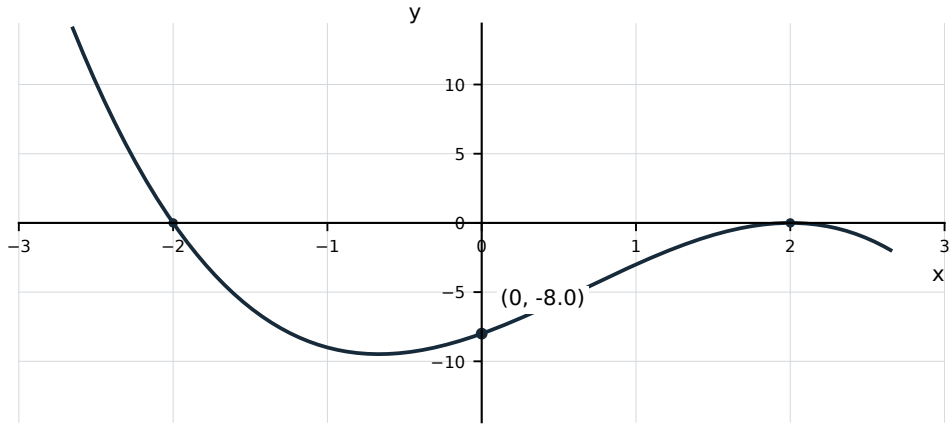
$A.\quad f(x) = x^2 + 5,\ g(x) = 4x + 4$

$B.\quad f(x) = 4x + 4,\ g(x) = x^2 + 5$

$C.\quad f(x) = x^2 + 4,\ g(x) = 4x + 5$

$D.\quad f(x) = 4x^2 + 5,\ g(x) = x + 4$

4. Write a polynomial of the least possible degree matching this graph and the labeled point. Give factored form.



5. Find all zeros: (a) list Rational Root Theorem candidates, then test only -2 and 2; (b) use polynomial long division by the confirmed linear factor; (c) use the quadratic formula on the quotient. Record all zeros.

$$p(x) = x^3 + 5x^2 + 7x + 2$$

6. Find (a) domain and range; (b) both intercepts; (c) behavior as x tends to infinity and as x approaches the finite domain endpoint from within the domain.

$$f(x) = 2\sqrt{x + 9} - 6$$

7. Evaluate exactly using logarithmic properties; show the property used. Here log means base 10.

(a) $\log(0.8) + \log(1250)$

(b) $\dfrac{\log_7(64)}{\log_7(16)}$

8. Find (a) domain and range; (b) vertical asymptote and x-intercept; (c) behavior at both ends of the domain.

$$f(x) = \log_3(x - 3) + 2$$

9. Solve the nonlinear system algebraically.

$$y = x^2$$
$$y = x + 2$$

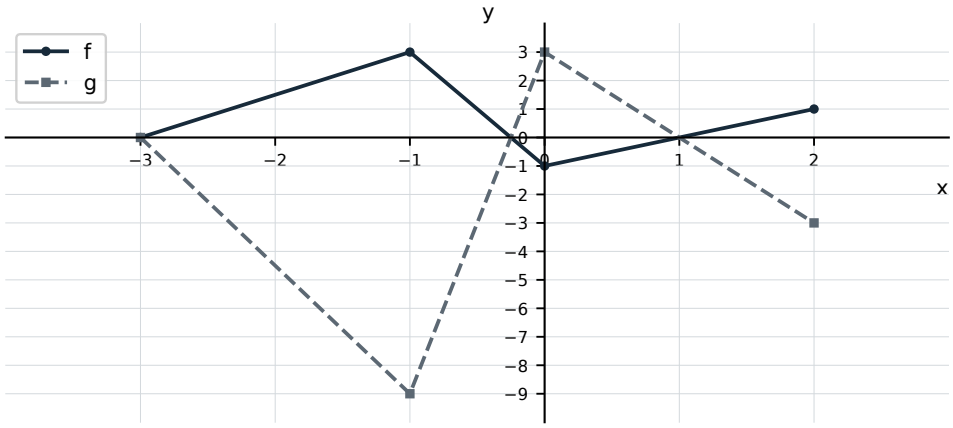
10. The solid graph is f and the dashed graph is g. Select the expression for g(x).

A. $g(x) = 3f(x)$

B. $g(x) = f(-3x)$

C. $g(x) = -3f(x)$

D. $g(x) = f(x) - 3$



Student answer sheet

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Question 1

(a) Domain	2 pts
(b) Hole	2 pts
(b) Vertical	1 pts
(b) Horizontal	1 pts
(c) x-intercept	2 pts
(c) y-intercept	2 pts

Question 2

(a)	3 pts
(b)	2 pts
(c)	3 pts
(d)	2 pts

Question 3

Choice	10 pts

Question 4

Polynomial	10 pts

Question 5

(a) Candidates/root	2 pts
(b) Quotient	4 pts
(c) Other zeros	4 pts

Student answer sheet

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Question 6

(a) Domain	2 pts
(a) Range	2 pts
(b) x-intercept	1 pts
(b) y-intercept	1 pts
(c) Infinite end	3 pts
(c) Finite end	1 pts

Question 7

(a)	5 pts
(b)	5 pts

Question 8

(a) Domain	2 pts
(a) Range	2 pts
(b) Asymptote	1 pts
(b) x-intercept	1 pts
(c) Left end	2 pts
(c) Right end	2 pts

Question 9

Intersections	10 pts

Question 10

Choice	10 pts

Answer key

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Version 202610122020.47 | Name: _____

Question 1

(a) Domain	$\mathbb{R} \setminus \{-3, 6\}$	2 pts
(b) Hole	$(-3,\ -\frac{1}{9})$	2 pts
(b) Vertical	$x = 6$	1 pts
(b) Horizontal	$y = 1$	1 pts
(c) x-intercept	$(-4,\ 0)$	2 pts
(c) y-intercept	$(0,\ -\frac{2}{3})$	2 pts

Question 2

(a)	C	3 pts
(b)	A	2 pts
(c)	D	3 pts
(d)	B	2 pts

Question 3

Choice	A	10 pts

Question 4

Polynomial	$-1(x+2)(x-2)^2$	10 pts

Question 5

(a) Candidates/root	$\{-2,\ -1, 1, 2\};\ x = -2$	2 pts
(b) Quotient	$x^2 + 3x + 1;\ R = 0$	4 pts
(c) Other zeros	$x = -\frac{3}{2} - \frac{\sqrt{5}}{2},\ \ -\frac{3}{2} + \frac{\sqrt{5}}{2}$	4 pts

Answer key

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Question 6

(a) Domain	$[-9, \infty)$	2 pts
(a) Range	$[-6, \infty)$	2 pts
(b) x-intercept	$(0,\ 0)$	1 pts
(b) y-intercept	$(0,\ 0)$	1 pts
(c) Infinite end	$x \to \infty :\ f(x) \to \infty$	3 pts
(c) Finite end	$x \to -9^{+} :\ f(x) \to -6$	1 pts

Question 7

(a)	3	5 pts
(b)	$\frac{3}{2}$	5 pts

Question 8

(a) Domain	$(3, \infty)$	2 pts
(a) Range	$(-\infty, \infty)$	2 pts
(b) Asymptote	$x = 3$	1 pts
(b) x-intercept	$(\frac{28}{9},\ 0)$	1 pts
(c) Left end	$x \to 3^{+} :\ f(x) \to -\infty$	2 pts
(c) Right end	$x \to \infty :\ f(x) \to \infty$	2 pts

Question 9

Intersections	$(-1,\ 1);\ (2,\ 4)$	10 pts

Question 10

Choice	C	10 pts